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Vehicle battery charging system using motor driving system

Abstract

A battery charging system includes a first inverter including first switching elements and DC and AC terminals connected to the battery and a first end of the windings of a motor, respectively; a second inverter including second switching elements and DC and AC terminals connected to the battery and a second end of the windings, respectively; third switching elements each having a first end connected to the second end of the windings and a second end connected to each other; a charging capacitor provided between the second end of the third switching elements and a negative terminal of the battery, and to which a DC charging voltage is applied for charging the battery; a fourth switching element connected in series with the charging capacitor between the second end of the third switching elements and the negative terminal; and a controller configured to control the first to fourth switching elements.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) The present application claims priority to Korean Patent Application No. 10-2022-0015595, filed on Feb. 7, 2022, the entire contents of which is incorporated herein for all purposes by this reference.

BACKGROUND OF THE PRESENT DISCLOSURE

Field of the Present Disclosure

(2) The present disclosure relates to a vehicle battery charging system using a motor driving system, and more specifically, to a vehicle battery charging system using a motor driving system, wherein a battery inside a vehicle may be charged by use of an open-end winding motor driving system which drives a motor by use of multiple inverters connected to both ends of windings inside the motor, respectively.

Description of Related Art

(3) The fuel efficiency (or energy efficiency) of an eco-friendly vehicle (e.g., electric car) which is powered by torque generated by a motor is determined by the power conversion efficiency of the inverter-motor, and maximizing the inverter power conversion efficiency and the motor efficiency is crucial to fuel efficiency improvement.

(4) Accordingly, a technology has been provided such that a closed-end winding mode in which first ends of windings inside a vehicle driving motor are connected to each other, forming a Y-connection, and the motor is then driven, or an open-end winding mode in which inverters are connected to both ends of motor windings, respectively, to drive the motor while both ends of the motor windings remain open, is selectively determined, driving the motor.

(5) Meanwhile, an electric vehicle (EV) or a plug-in hybrid electric vehicle (PHEV) commonly converts power provided by an external charging facility into a state appropriate for charging the battery inside the vehicle, and provides the same to the battery, charging the battery.

(6) Conventional charging facilities for fast charging are fabricated to output a single voltage specification of 400V, but batteries used inside vehicles tend to be designed to have a voltage of 800V or higher to improve the efficiency or the traveling distance. To charge batteries of vehicles having various voltage magnitudes, charging facilities capable of covering various ranges of voltages corresponding to battery voltage specifications need to be provided, or a single charging facility needs to be implemented to be able to output various ranges of voltages.

(7) There are problems in that huge costs are necessary to construct such a charging infrastructure, and if a low voltage needs to be output due to restrictions on the charging current imposed by the charging facility, the reduced charging power increases the charging time.

(8) Accordingly, the pertinent technical field requires a battery charging scheme, in connection with a motor driving system which drives a motor in an open-end winding mode by use of multiple inverters, capable of converting the magnitude of the charging voltage provided by a charging facility constructed as an existing infrastructure without an additional device and without additional cost increase, charging the battery.

(9) Furthermore, a charging capacitor is connected to a neutral terminal of a Y-connection of a motor to reduce the input voltage ripple of an external charger during fast charging and to maintain a stable voltage, and such a charging capacitor needs to be separated from the neutral terminal when a motor for traveling is driven. To the present end, a relay is applied on a charging current conducting path in a conventional motor driving system, but the large current (charging current) causes a problem of heating, degradation, or burning of the relay.

(10) The information included in this Background of the present disclosure is only for enhancement of understanding of the general background of the present disclosure and may not be taken as an acknowledgement or any form of suggestion that this information forms the prior art already known to a person skilled in the art.

BRIEF SUMMARY

(11) Various aspects of the present disclosure are directed to providing a vehicle battery charging system using a motor driving system, wherein the magnitude of a charging voltage provided by an external charging facility is appropriately converted by use of a motor driving system for driving a motor in an open-end winding mode without a separate dedicated conversion device, charging a battery.

(12) Furthermore, it is another aspect of the present disclosure to provide a vehicle battery charging system using a motor driving system, wherein a charging capacitor and a neutral point of a motor winding may be effectively connected to and disconnected from each other.

(13) Technical problems to be solved by the present disclosure are not limited to the above-mentioned technical problems, and other technical problems not mentioned herein will be clearly understood from the following description by those skilled in the art to which the present disclosure pertains.

(14) In accordance with an aspect of the present disclosure, a battery charging system for charging a battery using a motor driving system for driving a motor including a plurality of windings corresponding to a plurality of phases, respectively, may include: a first inverter including a plurality of first switching elements and having a direct current (DC) terminal connected to the battery and an alternative current (AC) terminal connected to a first end of the plurality of windings; a second inverter including a plurality of second switching elements and having a DC terminal connected to the battery and having an AC terminal connected to a second end of the plurality of windings; a plurality of third switching elements each having a first end connected to the second end of the plurality of windings and a second end connected to each other; a charging capacitor provided between the second end of the plurality of third switching elements and a negative terminal of the battery, and to which a DC charging voltage is applied in a charging mode for charging the battery; a fourth switching element connected in series with the charging capacitor between the second end of the plurality of third switching elements and the negative terminal of the battery; and a controller configured to turn on the fourth switching element in the charging mode and to control open/short-circuited states of the plurality of first switching elements, the plurality of second switching elements, and the plurality of third switching elements based on a magnitude of the DC charging voltage and a magnitude of a voltage of the battery.

(15) For example, the battery charging system may further include a first charging power application switch of which a first end is connected to the second end of the plurality of third switching elements and in which a high-potential of the DC charging voltage is applied to a second end thereof, and a second charging power application switch of which a first end is connected to the negative terminal of the battery and in which a low-potential of the DC charging voltage is applied to a second end thereof, and the controller may be configured to control the first charging power application switch and the second charging power application switch to be in the short-circuited state in the charging mode.

(16) For example, the controller may turn off the fourth switching element in a driving mode for driving the motor.

(17) For example, the charging capacitor may be electrically separated from a neutral terminal of the motor formed at the second end of the plurality of third switching elements when the fourth switching element is turned off.

(18) For example, the fourth switching element may include an FET.

(19) For example, the fourth switching element may be mounted on an inverter board on which the first inverter and the second inverter are mounted.

(20) For example, the charging capacitor may be provided between the second end of the plurality of third switching elements and the fourth switching element.

(21) For example, the fourth switching element may be provided between the second end of the plurality of third switching elements and the charging capacitor.

(22) For example, when the DC charging voltage is lower than the voltage of the battery, the

controller may, in the charging mode, control all of the plurality of third switching elements to be in the short-circuited state, and may perform pulse width modulation control on a lower switching element of the first inverter to boost the DC charging voltage, the lower switching element of the first inverter being a switching element connected to a low-potential terminal of the DC terminal among the switching elements included in the first inverter.

(23) For example, when the DC charging voltage corresponds to the voltage of the battery, the controller may, in the charging mode, control all of the plurality of third switching elements to be in the short-circuited state, and may control the charging current to be transmitted to the battery through an upper switching element of the second inverter, the upper switching element of the second inverter being a switching element connected to a high-potential terminal of the DC terminal among the switching elements included in the second inverter.

(24) According to the vehicle battery charging system using a motor driving system, a motor driving system provided for open-end winding driving of a motor is utilized to appropriately convert the magnitude of a charging voltage according to the magnitude of a charging voltage provided by an external charger, charging a battery.

(25) Accordingly, the vehicle battery charging system using a motor driving system makes it unnecessary to construct an additional infrastructure for an additional high-voltage charging facility in line with the high voltage of vehicle batteries so that no social cost for infrastructure construction will be incurred.

(26) In addition, the vehicle battery charging system using a motor driving system enables battery charging in response to various external charging voltages together with motor driving in a highly-efficient open-end winding type.

(27) Moreover, the vehicle battery charging system using a motor driving system, controls whether to connect a charging capacitor and a neutral point of a motor winding through a semiconductor switching element, and thus is excellent in terms of heating, cooling, resistance to impacts, and burning compared with a relay.

(28) The methods and apparatuses of the present disclosure have other features and advantages which will be apparent from or are set forth in more detail in the accompanying drawings, which are incorporated herein, and the following Detailed Description, which together serve to explain certain principles of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a circuit diagram illustrating a vehicle battery charging system using a motor driving system according to an exemplary embodiment of the present disclosure;

(2) FIG. 2 and FIG. 3 are diagrams illustrating an operating state of a vehicle battery charging system using a motor driving system according to an exemplary embodiment of the present disclosure;

(3) FIG. 4 is a circuit diagram illustrating a vehicle battery charging system using a motor driving system according to another aspect of an exemplary embodiment of the present disclosure; and

(4) FIG. 5 is a flowchart illustrating an example of a control process of a motor driving system according to an exemplary embodiment of the present disclosure.

(5) It may be understood that the appended drawings are not necessarily to scale, presenting a somewhat simplified representation of various features illustrative of the basic principles of the present disclosure. The specific design features of the present disclosure as included herein, including, for example, specific dimensions, orientations, locations, and shapes will be determined in part by the particularly intended application and use environment.

(6) In the figures, reference numbers refer to the same or equivalent parts of the present disclosure

throughout the several figures of the drawing.

DETAILED DESCRIPTION

(7) Reference will now be made in detail to various embodiments of the present disclosure(s), examples of which are illustrated in the accompanying drawings and described below. While the present disclosure(s) will be described in conjunction with exemplary embodiments of the present disclosure, it will be understood that the present description is not intended to limit the present disclosure(s) to those exemplary embodiments of the present disclosure. On the other hand, the present disclosure(s) is/are intended to cover not only the exemplary embodiments of the present disclosure, but also various alternatives, modifications, equivalents and other embodiments, which may be included within the spirit and scope of the present disclosure as defined by the appended claims.

(8) Hereinafter, a vehicle battery charging system using a motor driving system according to various embodiments of the present disclosure will be described in detail with reference to the accompanying drawings.

(9) FIG. 1 is a circuit diagram illustrating a vehicle battery charging system using a motor driving system according to an exemplary embodiment of the present disclosure.

(10) Referring to FIG. 1, a vehicle battery charging system using a motor driving system according to an exemplary embodiment of the present disclosure is a charging system that utilizes a motor driving system for supplying driving power to a motor **100** including a plurality of windings **L1**, **L2** and **L3** corresponding to a plurality of phases.

(11) First, the motor driving system may include a first inverter **10** including a plurality of first switching elements **S11**, **S12**, **S13**, **S14**, **S15** and **S16** and to be connected to one end of each of the windings of the motor **100**, a second inverter **20** including a plurality of second switching elements **S21**, **S22**, **S23**, **S24**, **S25** and **S26** and to be connected to the other end of each of the windings of the motor **100**, and a plurality of third switching elements **S31**, **S32** and **S33** having one end connected to the other end of each of the windings of the motor **100** and the other end short-circuited to each other.

(12) The first inverter **10** may have a DC terminal to which a DC voltage formed between a positive terminal and a negative terminal of the battery **200** is applied and an AC terminal connected to each of the windings **L1**, **L2** and **L3** of the motor **100**. Two nodes of the first inverter **10** to which the positive terminal and the negative terminal of the battery **200** are connected may be the DC terminal, and three nodes in which the first inverter **10** is connected to one end of each of the windings of the motor **100** may be the AC terminal.

(13) Similarly, the second inverter **20** may have a DC terminal selectively connectable to the DC terminal of the first inverter **10** and an AC terminal connected to each of the winding **L1**, **L2** and **L3** of the motor **100**. Two nodes including one node in which the second inverter **20** is selectively connectable to the DC terminal of the first inverter **10** may be the DC terminal, and three nodes in which the second inverter **20** is connected to the other end of each of the windings of the motor **100** may be the AC terminal.

(14) In a motor driving mode in which the motor generates power for driving a vehicle, the first inverter **10** and the second inverter **20** have a common DC terminal connected to the battery **200** to convert DC power stored in the battery **200** into three-phase AC power and provide the AC power to the motor **100**, or to convert, into DC, regenerative braking energy generated by the generation of regenerative braking torque of the motor **100** during regenerative braking and provide the DC to the battery **200**. The present conversion between DC power and AC power may be performed by pulse width modulation control of the plurality of first switching elements **S11**, **S12**, **S13**, **S14**, **S15** and **S16** and the plurality of second switching elements **S21**, **S22**, **S23**, **S24**, **S25** and **S26** respectively provided in the first inverter **10** and the second inverter **20**.

(15) The first inverter **10** may include a plurality of legs **11**, **12** and **13** to which a DC voltage formed in the DC terminal is applied. Each of the legs **11**, **12** and **13** may correspond to each of the

phases of the motor **100** to form an electrical connection.

(16) The first leg **11** may include two switching elements **S11** and **S12** connected in series between two nodes forming a DC terminal, and a connection node of the two switching elements **S11** and **S12** may be connected to one end of the winding **L1** of one phase in the motor **100** so that AC power corresponding to one of the plurality of phases of the motor **100** is input and output.

(17) Similarly, the second leg **12** may include two switching elements **S13** and **S14** connected in series between two nodes forming a DC terminal, and a connection node of the two switching elements **S13** and **S14** may be connected to one end of the winding **L2** of one phase in the motor **100** so that AC power corresponding to one of the plurality of phases of the motor **100** is input and output.

(18) Furthermore, the third leg **13** may include two switching elements **S15** and **S16** connected in series between two nodes forming a DC terminal, and a connection node of the two switching elements **S15** and **S16** may be connected to one end of the winding **L3** of one phase in the motor **100** so that AC power corresponding to one of the plurality of phases of the motor **100** is input and output.

(19) The second inverter **20** may also have a configuration similar to that of the first inverter **10**. The second inverter **20** may include a plurality of legs **21**, **22** and **23** to which a DC voltage of a DC terminal to which a DC voltage between two terminals of the battery **200** is applied is applied. Each of the legs **21**, **22** and **23** may correspond to the plurality of phases of the motor **100** to form an electrical connection.

(20) The first leg **21** may include two switching elements **S21** and **S22** connected in series between two nodes forming a DC terminal, and a connection node of the two switching elements **S21** and **S22** may be connected to the other end of the winding **L1** of one phase in the motor **100** so that AC power corresponding to one of the plurality of phases of the motor **100** is input and output.

(21) Similarly, the second leg **22** may include two switching elements **S23** and **S24** connected in series between two nodes forming a DC terminal, and a connection node of the two switching elements **S23** and **S24** may be connected to the other end of the winding **L2** of one phase in the motor **100** so that AC power corresponding to one of the plurality of phases of the motor **100** is input and output.

(22) In addition, the third leg **23** may include two switching elements **S25** and **S26** connected in series between two nodes forming a DC terminal, and a connection node of the two switching elements **S25** and **S26** may be connected to one end of the winding **L3** of one phase in the motor **100** so that AC power corresponding to one of the plurality of phases of the motor **100** is input and output.

(23) The first inverter **10** may be connected to one end of the windings **L1**, **L2** and **L3** of the motor **100** and the second inverter **20** may be connected to the other end of the windings **L1**, **L2** and **L3** of the motor **100**. That is, an electrical connection of an open-end winding structure in which both ends of the windings **L1**, **L2** and **L3** of the motor **100** are respectively connected to the first inverter **10** and the second inverter **20** may be formed.

(24) The switching elements **S11**, **S12**, **S13**, **S14**, **S15** and **S16** and **S21**, **S22**, **S23**, **S24**, **S25** and **S26** included in the first inverter **10** and the second inverter **20** may be switching elements included in a typical inverter for driving a motor, and should be understood as a concept including IGBT or FET in which switching is actually performed and a diode connected in the reverse direction between the source and the drain of the IGBT or FET.

(25) The plurality of third switching elements **S31**, **S32** and **S33** may be connected to the other end (AC terminal of the second inverter) of the windings **L1**, **L2** and **L3** of the motor **100**. The third switching elements **S31**, **S32** and **S33** may be referred to as “split switches”. One end of the plurality of split switches **S31**, **S32** and **S33** may be connected to one end of the windings **L1**, **L2** and **L3** of the motor **100**, and the other end thereof may be connected to each other to form an electrical short.

(26) When the plurality of split switches **S31**, **S32** and **S33** are opened, both ends of the windings **L1**, **L2** and **L3** of the motor **100** may be respectively connected to the first inverter **10** and the second inverter **20** to form an open-end winding structure. In addition, when the plurality of split switches **S31**, **S32** and **S33** are short-circuited, the other ends of the windings **L1**, **L2** and **L3** of the motor **100** may be electrically short-circuited to each other, and an electrical connection of a closed-end winding structure in which the windings **L1**, **L2** and **L3** of the motor **100** forms a Y-connection may be formed. When the electrical connection of the closed-end winding structure is formed, the motor **100** may be driven (i.e., CEW driving mode) by performing pulse width modulation control on the switching elements **S11**, **S12**, **S13**, **S14**, **S15** and **S16** of the first inverter **10**, and the switching elements **S21**, **S22**, **S23**, **S24**, **S25** and **S26** of the second inverter **20** may always maintain an open state.

(27) The plurality of split switches **S31**, **S32** and **S33** may be collectively referred to as a split switch unit **30** because they are configured to switch a circuit connection structure for driving the motor between the open-end winding structure and the closed-end winding structure.

(28) The controller **40** may perform pulse width modulation control on the switching elements **S11**, **S12**, **S13**, **S14**, **S15** and **S16** and **S21** to **S26** included in the first inverter **10** and the second inverter **20** so that the motor **100** may be driven based on an output required for the motor **100** in the motor driving mode.

(29) The controller **40** may determine an inverter used for driving the motor based on the required output of the motor **100** and accordingly, may determine the ON/OFF state of the plurality of split switches **S31**, **S32** and **S33** of the split switch unit **30**, so that it is possible to perform pulse width modulation control on the switching element of the converter whose driving is determined.

(30) For example, when the output required for the motor **100** is smaller than a predetermined reference value, the controller **40** may configure all of the plurality of split switches **S31**, **S32** and **S33** of the split switch unit **30** to a short-circuited state and may perform pulse width modulation control on the switching elements **S11**, **S12**, **S13**, **S14**, **S15** and **S16** of the first inverter **10** without operating the second inverter **20**, driving the motor **100** (i.e., a closed-end winding mode).

(31) The driving of the motor in the closed-end winding (CEW) mode may be achieved such that the controller **40** performs pulse width modulation control on the switching elements **S11**, **S12**, **S13**, **S14**, **S15** and **S16** of the first inverter **10** based on a DC voltage applied to the DC terminal of the first inverter **10**, a phase current provided to the motor **100** from the AC terminal of the inverter **10**, and a motor rotation angle detected by a motor rotor sensor provided in the motor **100**. Because various techniques for driving the motor **100** by performing pulse width modulation control on the plurality of switching elements in the inverter are already known in the art, further detailed description of the pulse width modulation control method of the inverter will be omitted.

(32) Meanwhile, when the output required for the motor **100** is greater than the predetermined reference value, the controller **100** may configure all of the plurality of split switches **S31**, **S32** and **S33** of the split switch unit **30** to the open state and may operate both the first inverter **10** and the second inverter **20** to drive the motor **100** (that is, the open-end winding mode). In the open-end winding mode, the motor **100** may be driven such that one end of the plurality of windings **L1**, **L2** and **L3** is in an open state and the other end thereof is also in an open state, and pulse width modulation control is performed on the two inverters **10** and **20** connected to both ends of the windings **L1**, **L2** and **L3**, respectively.

(33) The driving of the motor in the open-end winding mode may be achieved such that the controller **100** performs pulse width control on the first switching elements **S11**, **S12**, **S13**, **S14**, **S15** and **S16** of the first inverter **10** and the second switching elements **S21**, **S22**, **S23**, **S24**, **S25** and **S26** of the second inverter **20** together by receiving the DC voltages of the first inverter **10** and the second inverter **20**, the phase current provided to the plurality of windings corresponding to each phase of the motor **100**, the motor angle detected by a motor rotor sensor provided in the motor **100**, and the like.

(34) Various techniques for driving the motor by performing pulse width modulation control on the two inverters connected to both ends of the winding in the open-end winding mode are already known in the art, and thus a further detailed description thereof will be omitted.

(35) In a charging mode for charging the battery, the controller **40** may cause a DC charging voltage to be applied between the other end of the plurality of split switches **S31**, **S32** and **S33** and a negative terminal of the battery **200**, and may control the switching elements **S11**, **S12**, **S13**, **S14**, **S15** and **S16** and **S21** to **S26**, the switches **S31** to **S33**, and a fourth switching element **S4** included in the first inverter **10** and the second inverter **20** so that the magnitude of the DC charging voltage is converted and provided to the battery based on the magnitude of the DC charging voltage.

(36) The DC power provided from an external charger **300** may be applied between a node to which the plurality of split switches **S31**, **S32** and **S33** of the split switch unit **30** are interconnected and a node to which the negative terminal of the battery **200** is connected. That is, the charging voltage provided from the external charger **300** may be applied between a node to which the plurality of split switches **S31**, **S32** and **S33** are interconnected and a node to which the negative terminal of the battery **200** is connected.

(37) To form or block an electrical connection between the external charger **300** and the motor driving system, an exemplary embodiment of the present disclosure may further include charging power application switches **R1** and **R2** and the fourth switching element **S4**.

(38) The vehicle may include a charging inlet to which the external charger **300** is connected, and during charging, the charging inlet may be engaged with a charging outlet of the external charger **300**. The charging outlet of the external charger **300** may have a high-potential terminal (positive terminal) and a low-potential terminal (negative terminal) forming a charging voltage, and these terminals may be respectively connected to the high-potential terminal and the low-potential terminal of the charging inlet. A first charging power application switch **R1** may be connected between the high-potential terminal of the charging inlet and a node to which the plurality of split switches **S31**, **S32** and **S33** are interconnected, and a second charging power application switch **R2** may be connected between the low-potential terminal of the charging inlet and the negative terminal of the battery **200**.

(39) A charging capacitor **C** may be connected between the other end of the split switch unit **30** and the negative terminal of the battery. In addition, the fourth switching element **S4** may be connected in series with the charging capacitor **C**. In other words, one end of the charging capacitor **C** may be connected between the node to which the split switch unit **30** and the first charging power application switch **R1** are interconnected, and the other end of the charging capacitor **C** may be connected to the node to which the negative terminal of the battery **200** is connected.

(40) As described above, because the fourth switching element **S4** is connected in series with the charging capacitor **C**, it is not necessary to ensure performance assuming the conduction of a large current, unlike a case of using a relay on a path through which a large current (hundreds of A) of charging current conducts. Accordingly, it is possible to effectively control a connection between the charging capacitor **C** and a neutral terminal **N** even with a semiconductor type switching device without increasing the size or material cost for the large current. For example, a relatively small sized and inexpensive FET compared to the relay configured for conducting a large current may be applied to the fourth switching element **S4**. In the instant case, not only the material cost is reduced, but also an overall size of the inverter module is reduced, so that it is possible to achieve an effect of improving the output density. In addition, by mounting the FET for relay in an inverter board through PCB design, the FET for relay may be implemented without additional space, and the pin of the FET may be soldered to the board, so that a disadvantage of reliability deterioration due to vibration when driving the vehicle may be compensated. Additionally, by designing the location of the FET within a housing cooling range, there is an effect that the FET may also be cooled when a coolant is circulated for cooling the inverter board. However, because the FET is an exemplary embodiment of the fourth switching device **S4**, it is not necessarily limited thereto, and

it is apparent to those skilled in the art that various semiconductor-type switching devices may be applied.

(41) Therefore, in a case in which the external charger **300** is connected to the vehicle to perform charging, when the controller **40** turns on the split switch unit **30** and the fourth switching element **S4**, the neutral terminal N of the motor windings **L1**, **L2** and **L3** may be formed at the other end of the split switch unit **30**, and the charging capacitor **C** may be connected to the neutral terminal N. When the charging capacitor **C** is connected to the neutral terminal N, it is possible to reduce the input voltage ripple of the external charger **300** and stably maintain the voltage.

(42) However, the charging capacitor **C** needs to be electrically separated from the neutral terminal N for normal operation of the inverters **10** and **20** when the motor **100** is driven. Therefore, when the controller **40** turns off the fourth switching element **S4** at the time of driving of the motor **100**, the (-) terminal of the charging capacitor **C** and the ground may be separated, so that the charging capacitor **C** may be electrically separated from the neutral terminal N of the motor **100**.

(43) Meanwhile, the controller **40** may control the first charging power application switch **R1** and the second charging power application switch **R2** to be always open in the motor driving mode. In addition, the controller **40** may control the first charging power application switch **R1** and the second charging power application switch **R2** to always be in a short-circuited state in the battery charging mode.

(44) Various switching means known in the art may be applied to the first charging power application switch **R1** and the second charging power application switch **R2**, but the first charging power application switch **R1** and the second charging power application switch **R2** are used for maintaining the open/short-circuited state that are determined at the time of determining the mode switching or voltage rising/falling while the mode is continued, so that fast switching is not required. Accordingly, the first charging power application switch **R1** and the second charging power application switch **R2** are implemented as a relay.

(45) Here, the first charging power application switch **R1** and the second charging power application switch **R2** may be implemented in a form of a junction box **400** for forming an electrical connection relationship in a circuit. For example, the junction box **400** may be implemented in a form of hardware including a connection for forming an electrical connection between the high voltage terminal and the low voltage terminal of the battery **200** and the two terminals of the DC terminal of the inverter, a connection between the first and second charging power application switches **R1** and **R2** and the external charger, a connection between the first and second charging power application switches **R1** and **R2** with the split switch unit **30** and the DC terminals of the inverters **10** and **20**.

(46) FIG. 2 and FIG. 3 are diagrams illustrating an operating state of a vehicle battery charging system using a motor driving system according to an exemplary embodiment of the present disclosure.

(47) First, FIG. 2 illustrates an example in which the charging voltage (e.g., 400V) supplied from the external charger **300** in the charging mode is lower than the voltage (e.g., 800V) of the battery **200** of the vehicle.

(48) As shown in FIG. 2, the controller **40** may control all of the charging power application switches **R1** and **R2** to be in a short-circuited state in the charging mode, and may control switches **S31**, **S32**, and **S33** of the split switch unit **30** to maintain a short-circuited state. In addition, the controller **40** may control all of the switching elements **S21**, **S2**, **S23**, **S24**, **S25** and **S26** of the second inverter **20** to maintain an open state.

(49) Accordingly, windings **L1**, **L2** and **L3** of the motor **100**, diodes of upper switching elements **S11**, **S13**, and **S15** of the first inverter **10** connected to one end of the windings **L1**, **L2** and **L3**, and lower switching elements **S12**, **S14**, and **S16** of the first inverter **10** may form a topology of a boost converter configured for raising a voltage in a direction from the external charger **300** to the battery **200**. By use of the topology of the boost converter, the charging voltage provided from the external

charger **300** may be raised and provided to the battery **200** so that the battery **200** may be charged.

(50) In the instant case, because the voltage of the DC terminal of the second inverter **20** is greater than the voltage provided from the external charger **300**, a current flowing to the second inverter **20** may be blocked by the reverse diode of the upper switching elements **S21**, **S23**, and **S25** of the second inverter **20**.

(51) Here, the upper switching elements **S11**, **S13**, and **S15** refer to switching elements connected to the high-potential terminal of the DC terminal among the two switching elements included in the legs of the inverter, and the lower switching elements refer to switching elements connected to the low-potential terminal of the DC terminal.

(52) In addition, although FIG. 2 shows an example in which all the switches **S31**, **S32** and **S33** of the split switch unit **30** maintain the short-circuited state and then raise the voltage, the controller **40** may selectively cause only some of the switches **S31**, **S32** and **S33** of the split switch unit **30** to be short-circuited and then may perform pulse width modulation control on the lower switching element connected to the switch that maintains the short-circuited state, so that the voltage may be raised.

(53) In a case in which the switches **S31**, **S32** and **S33** of the split switch unit **30** corresponding to each phase of the motor maintain a short-circuited state at the time of raising the voltage and all of the lower switches **S12**, **S14**, and **S16** of the first inverter **10** are switched to raise the voltage, the controller **40** may control the boost converter corresponding to each phase in an interleaved manner.

(54) An interleaved control technique using the boost converter and plurality of boost converters, which include a coil (inductor), a diode, and a switch to increase the magnitude of the voltage through pulse width modulation control of the switch, is well known in the art, and thus an additional description thereof will be omitted.

(55) Next, FIG. 3 illustrates an example of a case (e.g., 800V) in which the charging voltage supplied from the external charger **300** corresponds to the voltage (e.g., 800V) of the battery **200** of the vehicle in the charging mode. In the instant case, there is no need to increase or decrease the charging voltage, and the charging current may be directly supplied to the battery **200** so that the battery **200** may be charged.

(56) As illustrated in FIG. 3, the controller **40** may control all of the charging power application switches **R1** and **R2** to be in a short-circuited state in the charging mode, and may control the switches **S31**, **S32**, and **S33** of the split switch unit **30** to maintain the short-circuited state. In addition, the controller **40** may control all of the switching elements **S11**, **S12**, **S13**, **S14**, **S15** and **S16** of the first inverter **10** to maintain an open state.

(57) Accordingly, the charging current may be transmitted to the battery **200** through the upper switches **S21**, **S23**, and **S25** of the second inverter **20**.

(58) FIG. 4 is a circuit diagram illustrating a vehicle battery charging system using a motor driving system according to another aspect of an exemplary embodiment of the present disclosure.

(59) In the motor driving system shown in FIG. 4 compared to the motor driving system illustrated in FIG. 1, the connection order of a charging capacitor **C'** and a fourth switching element **S4'** connected in series between the other end of the split switch unit **30** and the negative terminal of the battery is changed.

(60) Even through this change, in a same manner, whether the charging capacitor **C'** and the neutral terminal **N** are electrically connected to each other may be determined by the control of the fourth switching element **S4'** according to whether the battery is charged and the motor is driven, so that an appropriate element arrangement is possible according to the design of an inverter housing.

(61) FIG. 5 is a flowchart illustrating an example of a control process of a motor driving system according to an exemplary embodiment of the present disclosure.

(62) Referring to FIG. 5, first, in operation **S510**, the controller **40** may determine whether the current state of the vehicle is a motor driving mode or a fast charging mode.

(63) When the vehicle is currently in the motor driving mode, in operation S520A, the controller **40** may turn off the fourth switching element **S4** to electrically separate the neutral terminal N of the motor windings **L1**, **L2** and **L3** from the charging capacitor **C**.

(64) In addition, in operation S530, the controller **40** may determine a CEW driving mode or an OEW driving mode according to a requested output of the motor **100**, and may generate voltage and current commands for driving the motor according to the determined driving mode.

(65) In operation S540, based on the voltage and current commands, the controller **40** may control first, second, and third switching elements to drive the motor.

(66) On the other hand, when the vehicle is currently in the battery charging mode, in operation S520B, the controller **40** may turn on the fourth switching element **S4** to electrically connect the neutral terminal N of the motor windings **L1**, **L2** and **L3** with the charging capacitor **C**.

(67) The controller **40** may determine the charging voltage by recognizing a charger connection in operation S550, and may control the first, second, and third switching elements to raise the voltage so that the charging current may be transmitted to the battery or transmitted directly to the battery in operation S560, as described above with reference to FIG. 2 or 3.

(68) As described above, the vehicle battery charging system using the motor driving system according to various embodiments of the present disclosure may utilize the motor driving system provided for driving the open-end winding of the motor to appropriately convert the magnitude of the charging voltage according to the magnitude of the charging voltage provided from the external charger, facilitating the battery to be charged.

(69) Therefore, the vehicle battery charging system using the motor driving system according to various embodiments of the present disclosure may eliminate the need to build additional infrastructure for additional high-voltage charging facilities according to the high voltage of the vehicle battery, so that the social cost may be avoided.

(70) In addition, the vehicle battery charging system using the motor driving system according to various embodiments of the present disclosure may drive the motor in a high-efficiency open-end winding manner and may enable battery charging corresponding to various external charging voltages.

(71) In addition, because the charging current does not directly conduct to the fourth switching element, the fourth switching element may be implemented as a semiconductor-type switching element, so that the inverter module may be miniaturized, and cooling of the fourth switching element may be facilitated.

(72) In addition, the term related to a control device such as “controller”, “control apparatus”, “control unit”, “control device”, “control module”, or “server”, etc refers to a hardware device including a memory and a processor configured to execute one or more steps interpreted as an algorithm structure. The memory stores algorithm steps, and the processor executes the algorithm steps to perform one or more processes of a method in accordance with various exemplary embodiments of the present disclosure. The control device according to exemplary embodiments of the present disclosure may be implemented through a nonvolatile memory configured to store algorithms for controlling operation of various components of a vehicle or data about software commands for executing the algorithms, and a processor configured to perform operation to be described above using the data stored in the memory. The memory and the processor may be individual chips. Alternatively, the memory and the processor may be integrated in a single chip. The processor may be implemented as one or more processors. The processor may include various logic circuits and operation circuits, may process data according to a program provided from the memory, and may generate a control signal according to the processing result.

(73) The control device may be at least one microprocessor operated by a predetermined program which may include a series of commands for carrying out the method included in the aforementioned various exemplary embodiments of the present disclosure.

(74) The aforementioned invention can also be embodied as computer readable codes on a

computer readable recording medium. The computer readable recording medium is any data storage device that can store data which may be thereafter read by a computer system and store and execute program instructions which may be thereafter read by a computer system. Examples of the computer readable recording medium include Hard Disk Drive (HDD), solid state disk (SSD), silicon disk drive (SDD), read-only memory (ROM), random-access memory (RAM), CD-ROMs, magnetic tapes, floppy discs, optical data storage devices, etc and implementation as carrier waves (e.g., transmission over the Internet). Examples of the program instruction include machine language code such as those generated by a compiler, as well as high-level language code which may be executed by a computer using an interpreter or the like.

(75) In various exemplary embodiments of the present disclosure, each operation described above may be performed by a control device, and the control device may be configured by multiple control devices, or an integrated single control device.

(76) In various exemplary embodiments of the present disclosure, the control device may be implemented in a form of hardware or software, or may be implemented in a combination of hardware and software.

(77) In addition, the terms such as “unit”, “module”, etc. included in the specification mean units for processing at least one function or operation, which may be implemented by hardware, software, or a combination thereof.

(78) For convenience in explanation and accurate definition in the appended claims, the terms “upper”, “lower”, “inner”, “outer”, “up”, “down”, “upwards”, “downwards”, “front”, “rear”, “back”, “inside”, “outside”, “inwardly”, “outwardly”, “interior”, “exterior”, “internal”, “external”, “forwards”, and “backwards” are used to describe features of the exemplary embodiments with reference to the positions of such features as displayed in the figures. It will be further understood that the term “connect” or its derivatives refer both to direct and indirect connection.

(79) The foregoing descriptions of predetermined exemplary embodiments of the present disclosure have been presented for purposes of illustration and description. They are not intended to be exhaustive or to limit the present disclosure to the precise forms disclosed, and obviously many modifications and variations are possible in light of the above teachings. The exemplary embodiments were chosen and described to explain certain principles of the present disclosure and their practical application, to enable others skilled in the art to make and utilize various exemplary embodiments of the present disclosure, as well as various alternatives and modifications thereof. It is intended that the scope of the present disclosure be defined by the Claims appended hereto and their equivalents.

Claims

1. A battery charging system for charging a battery using a motor driving system for driving a motor including a plurality of windings corresponding to a plurality of phases, respectively, the battery charging system comprising: a first inverter including a plurality of first switching elements and including a direct current (DC) terminal connected to the battery and an alternative current (AC) terminal connected to a first end of the plurality of windings; a second inverter including a plurality of second switching elements and including a DC terminal connected to the battery and an AC terminal connected to a second end of the plurality of windings; a plurality of third switching elements each including a first end connected to the second end of the plurality of windings and a second end connected to each other; a charging capacitor provided between the second end of the plurality of third switching elements and a negative terminal of the battery, and to which a DC charging voltage is applied in a charging mode for charging the battery; a fourth switching element connected in series with the charging capacitor between the second end of the plurality of third switching elements and the negative terminal of the battery; and a controller configured to turn on the fourth switching element in the charging mode and to control open/short-circuited states of the

plurality of first switching elements, the plurality of second switching elements, and the plurality of third switching elements based on a magnitude of the DC charging voltage and a magnitude of a voltage of the battery.

2. The battery charging system of claim 1, further including: a first charging power application switch of which a first end is connected to the second end of the plurality of third switching elements and in which a high-potential of the DC charging voltage is applied to a second end thereof, and a second charging power application switch of which a first end is connected to the negative terminal of the battery and in which a low-potential of the DC charging voltage is applied to a second end thereof, wherein the controller is configured to control the first charging power application switch and the second charging power application switch to be in the short-circuited state in the charging mode.

3. The battery charging system of claim 1, wherein the controller is configured to turn off the fourth switching element in a driving mode for driving the motor.

4. The battery charging system of claim 3, wherein the charging capacitor is electrically separated from a neutral terminal of the motor formed at the second end of the plurality of third switching elements when the fourth switching element is turned off.

5. The battery charging system of claim 1, wherein the fourth switching element includes a Field Effect Transistor (FET).

6. The battery charging system of claim 1, wherein the fourth switching element is mounted on an inverter board on which the first inverter and the second inverter are mounted.

7. The battery charging system of claim 1, wherein the charging capacitor is provided between the second end of the plurality of third switching elements and the fourth switching element.

8. The battery charging system of claim 1, wherein the fourth switching element is provided between the second end of the plurality of third switching elements and the charging capacitor.

9. The battery charging system of claim 1, wherein, when the DC charging voltage is lower than the voltage of the battery, the controller, in the charging mode, is configured to control all of the plurality of third switching elements to be in the short-circuited state, and configured to perform pulse width modulation control on a lower switching element of the first inverter to boost the DC charging voltage, the lower switching element of the first inverter being a switching element connected to a low-potential terminal of the DC terminal among the switching elements included in the first inverter.

10. The battery charging system of claim 1, wherein, when the DC charging voltage corresponds to the voltage of the battery, the controller, in the charging mode, is configured to control all of the plurality of third switching elements to be in the short-circuited state, and configured to control the charging current to be transmitted to the battery through an upper switching element of the second inverter, the upper switching element of the second inverter being a switching element connected to a high-potential terminal of the DC terminal among the switching elements included in the second inverter.
